

Junye Zhou

Portfolio (moonlitjaderabbit.github.io) | linkedin.com/in/junye-zhou-5ab895367 | junye.zhou@mail.utoronto.ca

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering

2025 – 2030

- Relevant coursework: CIV102 (structures), PHY180 (classical mechanics lab), Praxis I & II (engineering design)
- CIV102 matboard bridge ranked **#1 in cohort**, **#2 in EngSci Class of 2025**

PROJECTS

CIV102 Matboard Bridge (Team 602) | *Python, SolidWorks, OptiCutter*

Nov 2026 – Dec 2026

- Designed and built a 1.25 m matboard bridge for train loading; predicted failure load ~ 954 N after safety margin
- Led hand calculations with teammates; contributed to Python structural analysis (GitHub)
- Built and tested full-scale prototype; splice failure near 1000 N drove four design iterations (diaphragms, splice reinforcement)
- Team: Declan Chan, Junye Zhou, Navneet Saxena, Tianhao Fu

Zipline Variable Resistance System (Team 7) | *Requirements, capstan design*

Feb 2026 – Apr 2026

- Partner: Mississauga Aquatic Club (MSSAC); RFP focus on variable swim resistance for training
- Developed capstan-based shuttle with 8 resistance levels (60–170 N) and ~ 3 s adjustment time
- Built requirement tables, material Pugh charts, and verification plans; presented at Praxis II showcase
- Team: Grant Liang, Sean Ding, Nathan Xia, Junye Zhou

STL Robotics Community RFP (Team 7) | *Stakeholder research, technical writing*

Jan 2025 – Feb 2025

- Team RFP for St. Theresa of Lisieux VEX robotics (STL Robotics): collect loose parts from portable floors and sort metallic hardware for reuse
- Conducted stakeholder interviews and portable-space measurements; framed safety and ergonomics requirements
- Team: Grant Liang, Sean Ding, Nathan Xia, Junye Zhou

PHY180 Damped Pendulum Lab | *OSP Tracker, Python, data analysis*

Oct. 2025 – Oct. 2025

- Solo lab report: filmed pendulum at 60 fps; tracked motion frame-by-frame in OSP Tracker
- Analyzed period vs. angle, damping, length scaling, and Q vs. length; $Q \approx 109$ (two methods agreed)
- Documented uncertainty sources and compared results to course theory

Praxis I — Bed Dome Cover | *Proxy testing, Pugh analysis*

Sept 2025 – Dec 2025

- Alpha Release design for conflicting sleep/study lighting in UofT Chestnut Residence double rooms
- Diverged on four concepts; ran proxy illuminance tests in a real dorm; converged with Pugh charts
- Recommended foldable bed dome cover targeting <1 lux at the sleeper's eyes
- Team: Jiaxu Zhao, Junye Zhou, Jocelyn Li, Audrey Lin

TECHNICAL SKILLS

Languages: Python, Java, C

Engineering tools: SolidWorks, OSP Tracker, OptiCutter

Design methods: Requirements (NGO), Pugh matrices, proxy testing, stakeholder interviews, concept selection